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Inner Speech Classification Using Deep Learning Techniques for EEG-Based Brain-Computer Interfaces

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Abstract

Brain Computer Interface (BCI) is a technology that enables direct brain-to-external device communication without the use of muscles. Inner speech refers to the silent process of mentally imagining or "speaking" to oneself without any vocal output, which is distinct from overt speech. Inner speech classification is an emerging BCI paradigm that seeks to decode these silent thoughts or imagined words from brain signals, typically captured via electroencephalogram signals (EEG). A significant difficulty for this type of BCI is low classification accuracy when dealing with multi-class tasks. This work talks about a full-scale study on the classification of inner speech using EEG signals. We exploit sophisticated deep learning techniques, involving long short-term memory (LSTM), bidirectional LSTM (BLSTM), Transformer, and an ensemble method via voting classifier, with the aim to explore the ability of these architectures in decoding inner speech. Our experiments were based on two benchmark datasets with four Classes: Thinking Out Loud and BCI 2a, involving different topics and situations. The proposed methodology achieved high classification accuracies of 98.81% and 90.58% with the transformer model on Thinking Out Loud and BCI 2a dataset respectively, which evidences the success of our approach. In addition, features generated by our models remarkably decrease intraclass variability while preserving vital inter-class differences, as illustrated by t-SNE visualizations. More specifically, the results show that the algorithm developed within this paper may be confidently applied for multi-class classification of InnerSpeech using EEG signals.

1 Introduction

Brain-Computer Interfaces (BCIs) enable direct communication between the brain and external devices, offering crucial assistance to individuals with severe motor disabilities. These systems translate neural signals into commands that can control prosthetics, communication devices, or other assistive technologies, bypassing the need for muscle activity. BCIs have gained prominence for their potential to restore lost functions and provide alternative communication pathways in cases where traditional methods are ineffective.

Traditional BCI paradigms, such as motor imagery and event-related potentials (ERPs), rely on the brain's response to imagined movements or external stimuli. While these methods have shown promise, they often require extensive training, depend on external stimuli, and suffer from low Information Transfer Rates (ITR). These limitations have prompted a shift toward more user-friendly and non-invasive techniques, particularly electroencephalography (EEG).

EEG signals, recorded through electrodes placed on the scalp, offers a safer and more accessible alternative to invasive techniques like electrocorticography (ECoG), where electrodes are implanted on the brain's surface. Despite its lower spatial resolution, EEG's ease of use and non-invasiveness make it a preferred choice for developing practical and widely applicable BCIs.

Inner speech refers to the internalised verbal expression that occurs within the mind without any vocalisation or movement of the speech muscles. It carries out a very important function in cognitive processes such as decision-making and self -regulation. In BCIs, inner speech poses a very unique challenge and opportunity because it is fundamentally different from imagined speech, which involves mentally simulating speech processes with articulatory movements, and silent speech, which actually involves movement of speech muscles without sound production. Inner speech decoding from EEG signals offers more intuitive, brain-driven communication that goes well beyond the use of exogenous stimuli or physical articulation in order to achieve and advance smoother BCI systems.

Perhaps the biggest challenge to this task of classifying inner speech in EEG is extreme variability in EEG signals. Models developed for one person do not work very well for most others due to uniqueness of brain activity in each person. This variability is compounded by the fact that the signal-to-noise ratio is relatively low in EEG recordings; the relatively weak neural signals associated with inner speech are often drowned out by background noise and simply other electrical activity in the brain. That is why attainment of a reliable and consistent classification over a wide range of subjects is still quite a formidable challenge.

Our work complements other research in the field of brain-computer BCIs by concentrating on a difficult task, namely inner speech classification. Unlike vocalization or idea kinesthetic, innerspeech is subtle and more difficult to discern. Innerpeech is the spoken, inner verbal expression that comes with great special features. Problems in decoding of EEG signals will be emerging as significant sub-challenges for signal processing. We address this challenge by using deep learning approaches.There are three types of recurrent neural networks, which are designed as Short-Term Memory (LSTM), Bidirectional LSTM (BLSTM), and Transformer models to decode inner speech signals to comprehend speech signals originating from individual transformers. These models, each with its own advantages in modeling the temporal dependencies and interconnected sets of anatomical neural networks are collectively wired into a group system. The ensemble uses a voting classifier and weighted averaging to aggregate the estimations, to improve the general reliability and reliability of the proposed classification method for unbalanced text data. Our work pushes the boundary of inner speech decoding, presenting a framework that is not only innovative but also adaptable for different subjects, considering the high variability in EEG signals.

The paper is structured as follows: the next section reviews the related works, outlining the current state of research in inner speech classification and EEG-based BCIs. Following this, the methodology section describes the dataset used, detailing how EEG data was recorded and preprocessed for this task. The proposed methods section provides an in-depth explanation of the deep learning models utilized, including the LSTM, BLSTM, and Transformer architectures, and how the ensemble system is designed. The results section presents the performance evaluation of the models, and the paper concludes with a discussion of the findings and their implications for future BCI development.

2 Related Works

The integration of machine learning and deep learning techniques with EEG signal processing has significantly advanced healthcare, particularly in neuroimaging and diagnostics. EEG, a critical tool for monitoring brain activity, has seen extensive application in health-related studies, where machine learning classifiers like Naive-Bayes, Decision Tree, Random Forest, and Support Vector Machine (SVM) have been used to enhance classification accuracy as described by Hosseini et al. [1]. In addition, application of deep learning methods has has also been reported and widely discussed by Xu and Xia [10]. Research [2] proves that other forms of supervised learning techniques like SVM and K-Nearest that are preferred today were not very efficient for the task as well. K-Nearest Neighbors (KNN), in the main, are more accurate than any other algorithms especially when used with dimensionality reduction techniques and ensemble methods together. However, Vishwanath et al. [3] has further generalized these techniques to diagnosis systems such as epilepsy detection, schizophrenia classification, and with Traumatic Brain Injury (TBI) diagnosis, showing the role of machine learning in the potential of increasing the frequency and for increasing the efficiency of EEG-based diagnostics.

In the study by Sarmiento and his colleagues [11], the authors have proposed a deep learning algorithm called CNNeeg1-1 that was developed to recognize imagined vowel tasks using EEG signals. The performance of this algorithm was tested using both an open-access invaluable database and a novel dataset created in the study, indicating better outcome or result for most intra-subject training, especially with the newly developed dataset it got an accuracy of 85.66%. In addition, work published in [12] developed a multimodal database using concurrent EEG, facial, and audio data during imagined and vocalized phonemic prompts, providing a great demonstration of the effectiveness of deep-belief networks. Phonological classification activities analysis showed that the EGAM successfully represented distinctions in a typical phonological classification task. This multimodal approach recorded above 90% in the classification of consonants and above 95% in distinguishing one form of AV recording from another, and exposing some of the possible areas for further development of an advanced intelligent silent speech and brain computer interface technology. Extending the usage of EEG classification, Mahmood et al. [5] were successful at overcoming the problems of low accuracy and high computational complexity. In multi-class motor imagery classification within Brain- Computer interfaces, the implications of their innovative approach of Common Spatial are the following: Common Spatial Pattern (CSP) and SVM focussed on mu and beta rhythms for feature extraction. This classification, was applied on on the BCI Competition III dataset IIIa,and achieved an accuracy of 85.5% while cutting the number of computations.

Further advancements in EEG classification have explored deep learning techniques, particularly in the context of neuromuscular disorders. Zare and Sun [6] proposed a novel approach integrating Convolutional Neural

Networks (CNNs) with a transformer encoder block, aiming to capture the spatial-temporal features of raw EEG signals more effectively. Preliminary results from this study show promising performance, indicating the potential of this integrated model to advance motor imagery classification for BCI applications. Research work [7], employed Singular Value Decomposition (SVD) to suppress artifacts and capture channel-specific characteristics, while a deep Boltzmann machine was used to model these characteristics for motor imagery classification. Their method achieved a 14.2% higher classification accuracy compared to the common spatial pattern technique, with results validated by statistical analysis showing a significant difference ($p < 0.05$).

Along with these developments, van den Berg et al.'s recent work [8] has explored the use of EEG signals for the classification of inner speech. Their research delves into a new BCI paradigm that centers on the internalized process of voice imagination. This work sought to categorize EEG signals from eight people thinking about four different words using a 2D CNN based on the EEGNet architecture. While the obtained average word recognition accuracy of 29.7% is only marginally better than random, it is a significant step toward the development of more advanced techniques for inner speech classification and identifies areas that require additional study and improvement. Gasparini et al. [9] utilized the 'Thinking Out Loud' dataset and the 'Imagined Speech Dataset' to explore this area further. Their investigation revealed that deep learning models—specifically, LSTM and BLSTM—performed better than ensemble approaches like XGBoost and simple methods like SVM, even though both types of methods produced respectable outcomes. The highest average accuracy of 36.12% was achieved by BLSTM due to its capacity to capture sequential dependencies in both directions.

The amalgamation of sophisticated machine learning and deep learning methodologies has markedly improved EEG categorization in diverse applications. By examining deep learning ensemble methods for inner speech classification, our work advances this subject. This investigation highlights the possibility of integrating various deep learning techniques to enhance real-world neurotechnology applications.

3 Methodology

3.1 Dataset

The proposed strategies were tested on two publicly available datasets - the Thinking Out Loud and BCI 2a datasets.

3.1.1 Thinking Out Loud

The "Thinking Out Loud" dataset [13] was utilized for the analyses in this study, focusing on an inner speech paradigm aimed at controlling a Brain-Computer Interface (BCI) system through the imagination of specific Spanish words.

The experiment focused on a natural BCI control application using the Spanish words "arriba" (up), "abajo" (down), "derecha" (right), and "izquierda" (left). Each participant underwent three consecutive sessions, with 200 words per session, separated by short breaks. Each session included a baseline recording (15 seconds), a pronounced speech run, two inner speech runs, and two visualized condition runs. During the inner speech condition, participants imagined their own voice repeating a given word. Each trial began with a white circle on the screen, followed by a white triangle pointing in one of the four directions.Participants performed the indicated task when the triangle disappeared and the circle reappeared, stopping when the circle turned blue. Attention was assessed by randomly asking participants to recall the last inner speech and visualized conditions, with responses given via keyboard arrows. Fig.1 illustrates the structure of each inner speech trial analyzed in this paper.

Data acquisition was performed using EEG helmet with 128 wet electrodes and 8 externally active EOG/EMG wet electrodes, with a 24-bit resolution and a sampling rate of 1024 Hz. The EEG signals from the "Thinking Out Loud" dataset were preprocessed, using a band-pass filter between 0.5-100 Hz, a notch filter at 50 Hz, and downsampling to 254 Hz.

3.1.2 BCI 2a

The BCI Competition II dataset 2a [21] includes EEG recordings of four distinct motor imagery tasks imagining movement of the left hand, right hand, both bases, and lingo. For each subject, two sessions were recorded on different days. Each session consisted of six runs, with short breaks in between. Each run included 48 trials(12 per class), performing in an aggregate of 288 trials per session. At the launch of each trial, a cross appeared on a black screen that is fixed, accompanied by a short aural warning tone. After two seconds, a cue representing left, right, down, or over(corresponding to the left hand, right hand, foot, and tongue) appeared and remained on the screen for 1.25 seconds, motioning the subject to perform the corresponding motor imagery task. No feedback was handed during the trials, and actors were instructed to continue the motor imagery task until the obsession cross faded at t = 6 s. Fig.2 illustrates the recording paradigm anatomized in this paper. EEG signals were recorded

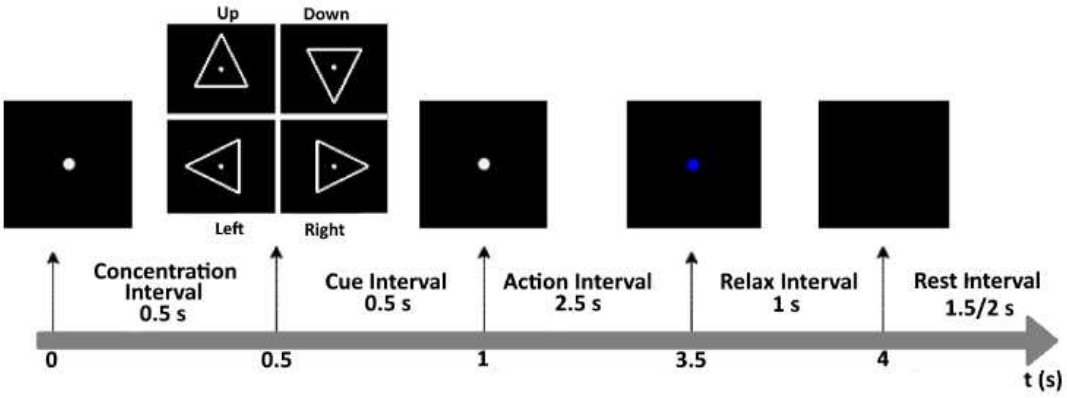


Figure 1 Structure of Inner speech trial outlined in the "Thinking Out Loud" dataset paper [13].

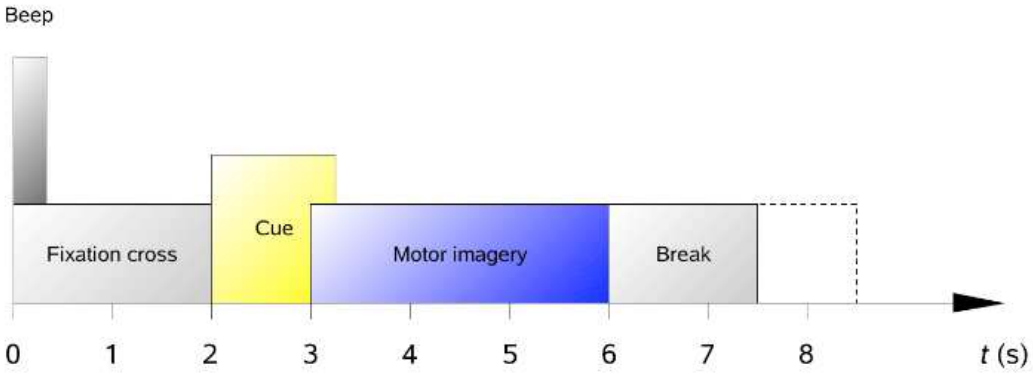


Figure 2 Structure of Inner speech trial outlined in the BCI 2a dataset [21].

using 22 Ag/ AgCl electrodes, with inter-electrode distances of 3.5 cm. The signals were tried at 250 Hz and bandpass filtered between 0.5 Hz and 100 Hz. Also, a 50 Hz notch sludge was applied to suppress line noise, and the amplifier perceptivity was set to 1 mV.

In addition to existing BCI and thinking-out-loud datasets, we also designed and recorded our own inner speech EEG data using the Emotiv Insight headset, which provides five channels at a sampling frequency of 128 Hz. The experiment was structured into distinct phases, including fixation, preparation, instruction, and simulation, with signals collected during the simulation phase where participants silently imagined directional commands—left, right, up, and down. The recorded signals were preprocessed through baseline correction, event isolation, high-pass filtering, and independent component analysis (ICA) to remove artifacts. From these cleaned signals, relevant features were extracted and used to train classification models, enabling the system to recognize imagined directions directly from EEG activity. This custom dataset allowed us to validate model performance on real-world data beyond publicly available benchmarks.

3.2 Proposed Methods

The EEG signals obtained from the dataset were first normalized as part of the preprocessing stage. Each normalized signal was then fed into LSTM, BLSTM, and Transformer models. LSTM, BLSTM, and Transformer models for EEG inner speech classification leverages their complementary strengths in handling temporal and contextual information. The results from these models were integrated, and was assigned to one of the four classes, using the ensemble method. This in turn improves the performance of the model and also increases its reliability. The block diagram of our proposed methodology is shown in Fig.3 below.

3.2.1 LSTM

In this study, we utilized Long Short-Term Memory (LSTM) networks to classify EEG signals associated with inner speech tasks. LSTM networks are best for handling seqential data because they are good at recognizing temporal dependencies that exist in time dependent EEG signals. For LSTM structure, the cell states are of memories that

have different durations and operate an input gate, a forget gate, and an output gate all of which employ the sigmoid function. The input gate which controls the levels of input data to be stored on the memory is illustrated in (1). The tanh in eqn. 2 aids formation of the memory state Ct to favor transfer of values only within the range 0 and 1.

$$i_t = \sigma[W_{ih}h_{t-1}, x_t] + b_i) \tag{1}$$

$$C_t = \tanh(W_{ch}h_{t-1}, x_t + b_c) \tag{2}$$

Then the forget gate decides which data to discard, based on the sigmoid function. It subtracts a value between 0 (discard) and 1 (retain) for each element in the memory cell, as described in (3).

$$C_{t-1}f_t = (W_F h_{t-1}, x_t + b_f) \tag{3}$$

The output gate represents the output of the LSTM. The output gate can be calculated as in step (4) using the sigmoid function, and the final hidden state is obtained in step (5) by combining the output gate with the hyperbolic tangent of the memory cell state, C_t.

$$O_t = \sigma(W_{oh}h_{t-1}, x_t + b_o) \tag{4}$$

$$h_t = O_t * tanh(C_t) \tag{5}$$

The final classification layer uses a softmax activation to classify the EEG signals into one of the four inner speech classes. The model was trained with early stopping and checkpointing mechanisms to optimize performance while avoiding overfitting.

3.2.2 BLSTM

BLSTMs process data in both forward and backward directions, enhancing the model's ability to grasp complex patterns and transitions by considering past and future contexts.Each element of an input sequence also computes the forward and reverse of the given input sequence, generating a hidden sequence in the forward direction and another running upwards in the backward direction. The architecture consists of two LSTM layers operating in a bidirectional manner, which is an important requirement for the model. The model preserves information from past as well as future experiences. This bidirectional approach is particularly effective for processing sequential data, where context from both directions can significantly improve classification accuracy.

3.2.3 Transformer

The Transformer model operates through the integration of fully-connected layers with self-attention mechanisms. That is, with a point-wise, fully connected layers with layered self-attention mechanism within an encoder-decoder architecture. The input is first received by a model passed to an encoder input layer with linear layer which has the size of dimension d_model parameter to transform the input into a more complex space, and then.

Positional encoding is incorporated into the input embeddings to convey the order of each token within the sequence. This encoding uses sine and cosine functions to represent relative positions, with even and odd time points, as represented in (6) and (7).

$$PE_{(pos,2t)} = sin(\frac{pos}{1000^{\frac{2t}{d}}}) \tag{6}$$

$$PE_{(pos,2t+1)} = cos(\frac{pos}{1000^{\frac{2t+1}{d}}}) \tag{7}$$

Where t is the time point, d represents the dimension of the vector, and pos is the channel position.

To improve the model's capability to focus on different sections of the input concurrently, a multi-head attention mechanism is applied, composed of multiple layers of Scaled Dot-Product Attention.The model was able to calculate the queries vector (Q), the values vector (V), and the keys vector (K) by multiplying the input vector by three different weight matrices as represented in (8).The Multi-Head Attention mechanism is integral in recognizing intricate data patterns.

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \tag{8}$$

The Transformer at its core is divided into two major submodules: the first comprises of a multi-head attention layer accompanied by a normalization layer and the second layer consist of the fully connected feed-forward layer, then a normalization layer is implemented. Residual connections around each submodule is also ensured, as well for the preservation of key data and enhance the performance of the model in classification of the EEG signals associated with inner speech and most of other as evaluated in other studies.

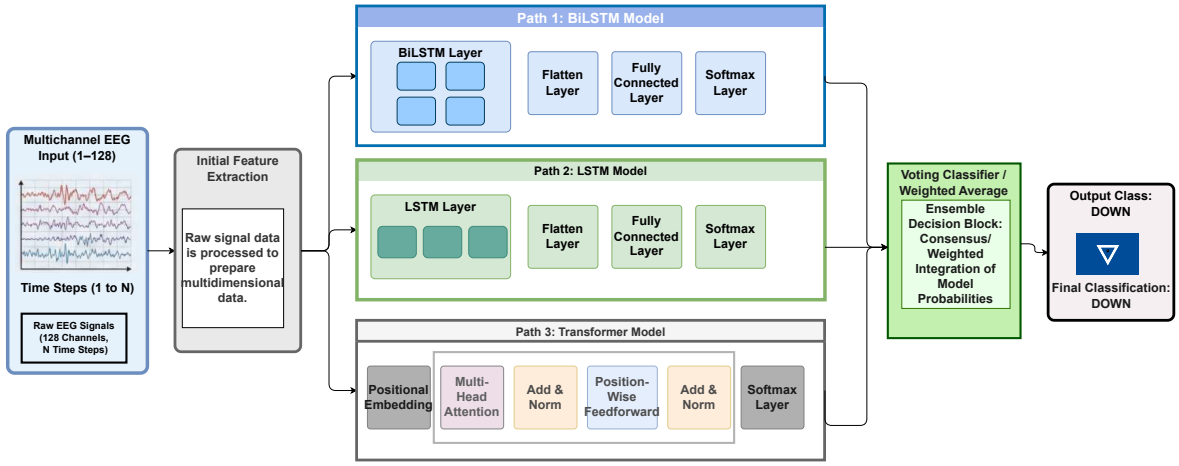


Figure 3 Block Diagram representing the processing of the signal by each model and the final output obtained using Ensemble technique

3.2.4 Ensemble Learning

After utilizing LSTM, BLSTM, and Transformer models for EEG signal classification, a hard voting classifier from ensemble learning is employed to improve overall performance. Hard voting integrates the output of each model by taking a majority vote. The last probability used for the class prediction corresponds to nature of the class labels used by the individual models. Among the important aspects of the results from each model is the nature of the class labels produced, that is, improving the classification capability by building on the advantages of each model. The hard voting classifier is mathematically represented as (9):

$$y_{final} = mode(y_1, y_2, ..., y_n) \tag{9}$$

where y_{final} is the final predicted class label, and $y_1, \ldots, y_n$ are the predicted labels from the individual models. This approach helps in reducing the variance of predictions, making the classifier more robust and accurate in distinguishing between the four classes of EEG signals related to inner speech.

Besides hard voting, we also used a weighted average ensemble, where every model is endowed with a distinctive weight depending on the performance of the individual model. The Transformer model, due to its higher capability of estimating dependency between temporal and spatial data. was assigned the maximum importance and was followed by BLSTM, LSTM. The weighted average ensemble method provides the prediction from each model and averages these predictions. The final prediction is determined by taking a weighted sum of the predicted probabilities or labels from each model.Mathematically, the formula for the weighted average classifier is given by (10):

$$y_{final} = w_1.y_1 + w_2.y_2 + ... + w_n.y_n \tag{10}$$

where y_{final} is the final predicted class label, $y_1, \ldots, y_n$ are the predicted labels from the individual models and $w_1, \ldots, w_n$ are the weights assigned to each prediction.This weighted approach improves classification accuracy by leveraging the strengths of each model while accounting for their differences in performance.

4 RESULTS AND DISCUSSION

Finally, in the results section, the performance of our EEG inner speech classification models using several key metrics: accuracy, precision, specificity, and F1 score. These metrics provide an analysis of the model in terms of its capacity to assign suitable classification labels for the EEG Signals and achieve a balance between true positives and true negatives against false positives and false negatives.

We have developed two distinct models for this study: a subject-independent model and a subject-dependent model.The subject-independent model was trained on all obtained data of subjects, emphasizing generalized patterns that apply to the whole set of the collected data. Table I and Table II display the results of the experiment of the subject-independent model across both datasets.

The accuracy metrics between the Thinking Out Loud and BCI 2a datasets differ due to variations in preprocessing quality. The Thinking Out Loud dataset is specifically designed for research, offering superior preprocessing, which contributes to better performance metrics. However, the Transformer model's ability to effectively

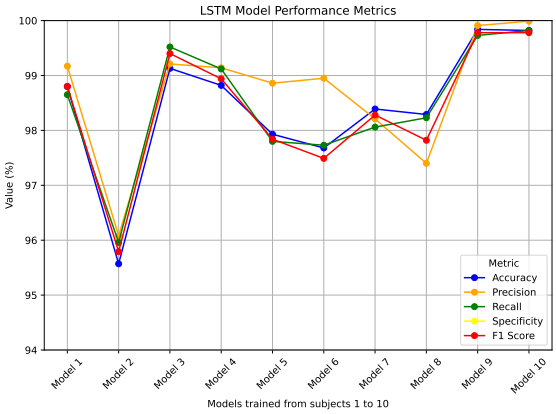


Figure 4: Performance Metrics of LSTM Subject Dependent Model - Thinking Out Loud Dataset

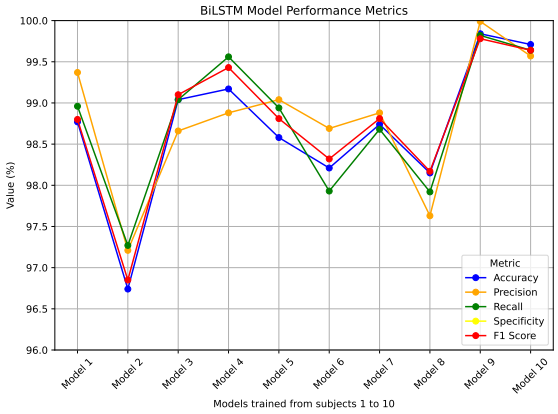


Figure 5: Performance Metrics of BLSTM Subject Dependent Model - Thinking Out Loud Dataset

capture patterns within the timestamps of EEG signals results in improved accuracy for the BCI dataset despite its less rigorous preprocessing.

Table 1 SUBJECT INDEPENDENT SCORES - Thinking Out Loud Dataset

MODEL	ACCURACY (in %)	PRECISION (in %)	RECALL (in %)	SPECIFICITY (in %)	F1 SCORE (in %)
LSTM	91.29	91.30	91.29	97.10	91.29
BLSTM	93.38	93.84	93.84	97.95	93.84
TRANSFORMER	96.43	96.44	96.45	99.65	98.19
VOTING CLASSIFIER	96.54	96.38	96.25	99.67	98.27

Table 2 SUBJECT INDEPENDENT SCORES - BCI 2a Dataset

MODEL	ACCURACY (in %)	PRECISION (in %)	RECALL (in %)	SPECIFICITY (in %)	F1 SCORE (in %)
LSTM	58.12	58.06	58.08	68.62	58.02
BLSTM	54.83	55.56	54.84	84.95	54.42
TRANSFORMER	90.53	97.80	87.09	99.65	93.07
VOTING CLASSIFIER	90.23	96.89	88.01	98.23	92.78

In contrast to the subject-independent models, the subject-dependent models were further fine-tuned for each individual subject, retraining the model on each subject's data to enhance accuracy. This method allows for a more personalized approach, potentially yielding better performance on an individual level by adapting to each subject's unique EEG signal patterns.

The results of the subject-dependent models using LSTM, BiLSTM, and Transformer architectures for the Thinking Out Loud dataset are depicted in Figures (4), (5) and (6). Model 9 achieved the highest accuracy of 99.84% with both LSTM and BiLSTM, while Model 3 reached an accuracy of 99.87% with the Transformer.

Similar training of models was also performed with the BCI 2a Dataset and the results are depicted in Fig (7), (8) and (9). Model 9 achieved the highest accuracy of 91.75% and 90.95 with LSTM and BiLSTM, while Model 9 reached an accuracy of 93.81% with the Transformer.

The Transformer model outperformed both the LSTM and BLSTM models on both datasets. LSTM, BLSTM and Transformer models have an average accuracy of 98.42, 98.69 and 98.81 with the Thinking Out Loud Dataset and an average accuracy of 88.78, 89.67 and 90.58 with the BCI 2a Dataset. The average accuracy showed improvement when transitioning from the subject-independent model to the subject-dependent model, highlighting the effectiveness of individual subject-specific training.

The Confusion Matrices (formed using Thinking Out Loud Dataset) show in Fig (10), (11) and (12), also help in analyzing the performance of these models and it can be seen that the transformer model outperformes LSTM and BLSTM.

In addition to evaluating individual models, we employed ensemble learning techniques using voting and

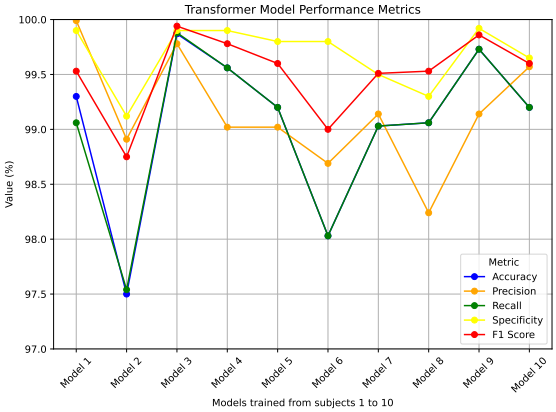


Figure 6: Performance Metrics of Transformer Subject Dependent Model - Thinking Out Loud Dataset

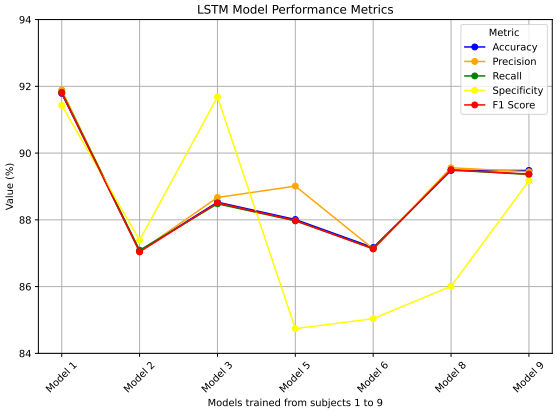


Figure 7: Performance Metrics of LSTM Subject Dependent Model - BCI Dataset

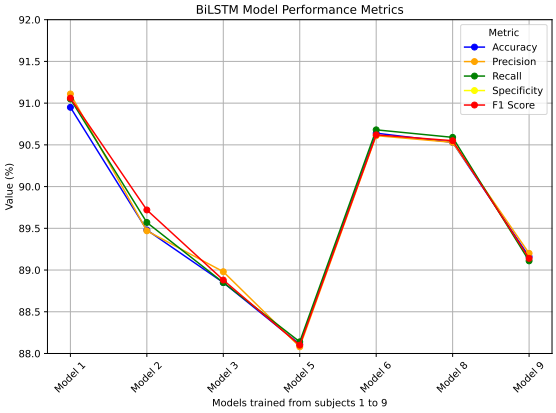


Figure 8: Performance Metrics of BLSTM Subject Dependent Model - BCI Dataset

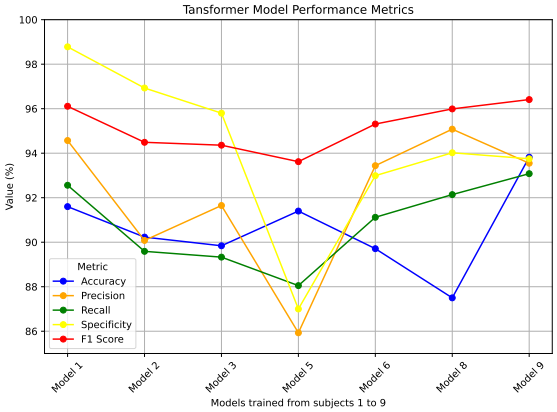


Figure 9: Performance Metrics of Transformer Subject Dependent Model - BCI Dataset

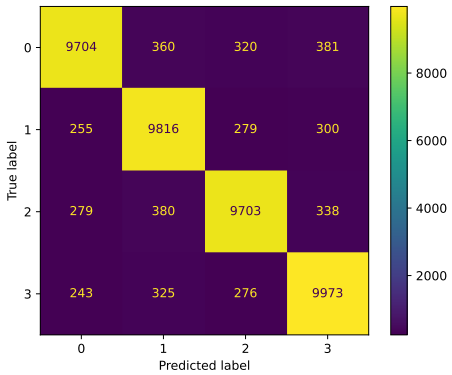


Figure 10: LSTM - Confusion Matrix

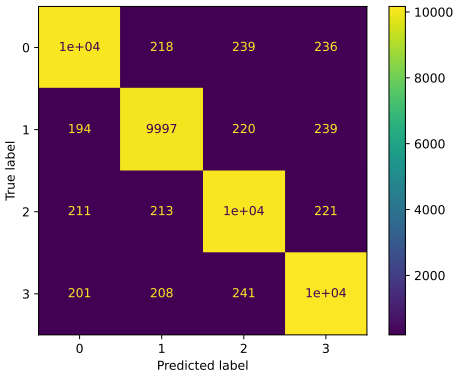


Figure 11: BLSTM - Confusion Matrix

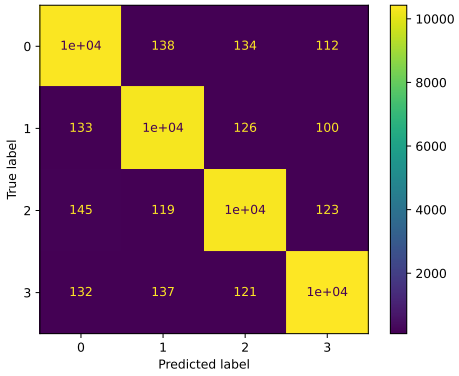


Figure 12: Transformer - Confusion Matrix

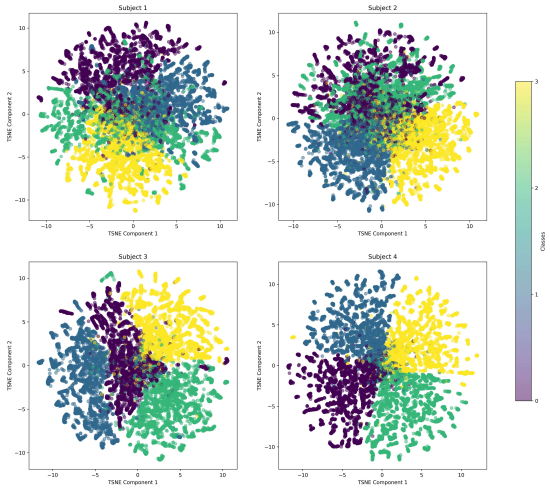


Figure 13: t-SNE Plot representing the classification by LSTM model

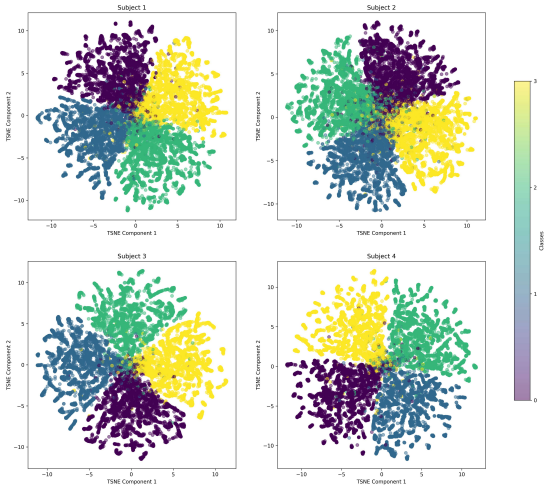


Figure 14: t-SNE Plot representing the classification by BLSTM model

weighted averaging with bagging to further enhance performance. Since each model demonstrated strong individual performance, ensemble methods provided valuable insights into their collective behavior.

For the voting-based ensemble, where the final prediction was determined by a majority vote among LSTM, BLSTM, and Transformer models, the result was predominantly influenced by the LSTM and BLSTM models, leading to an accuracy of around 91%. In contrast, the bagging ensemble, which incorporated weighted averaging, allowed the Transformer model to dominate the final predictions, yielding a higher accuracy of 96%.

The voting approach showed the influence of LSTM and BLSTM models, achieving solid but slightly lower accuracy, while the bagging method, which allowed the Transformer model to have more impact, produced notably higher results. This suggests that ensemble learning, particularly when weighted towards stronger models like the Transformer, can serve as a useful complementary approach to improve classification outcomes without relying exclusively on a single model.

Despite the subject-independent models for LSTM, BLSTM, and Transformer showing good accuracy around 91.29, 93.38, 96.43 respectively, testing these models on new sets of subjects—ones they were not trained on—results in accuracy dropping to around 25%. This decline is attributed to the uniqueness of EEG signals, making it challenging for the models to generalize across different subjects.The t-SNE plots as shown in Fig (13), (14) and (15) are of four different subjects classified by the same LSTM, BLSTM and Transformer models respectively. The difference in the classification pattern demonstrates the uniqueness of EEG signals for each subject. These plots reveal distinct clustering patterns for the four different classes, with clear separation observed across different subjects. This distinctiveness in patterns supports the idea that EEG signals vary significantly between individuals, further explaining the challenges in achieving high accuracy with subject-independent models.

In Table 3, the response time for each model, LSTM, BLSTM, and Transformer, is presented, illustrating the computational efficiency of the models when processing a single time step of EEG data. Table 4 compares the accuracy of models used for inner speech classification. The results from Gasparini et al [9]. are contrasted with

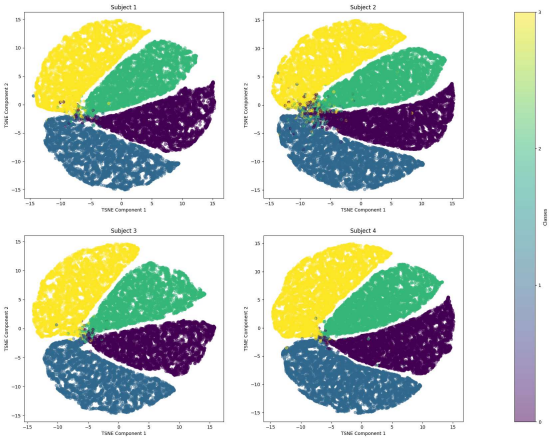


Figure 15: t-SNE Plot representing the classification by Transformer model

Table 3 Response time of each model for a single time step

Model	Response Time
LSTM	1 s
BiLSTM	1 s
Transformer	1.2 s

Table 4 Comparison with other works

Model	Gasparini et al. [9] (Accuracy %)	Proposed (Accuracy %)
LSTM	30.40	91.29
BiLSTM	36.10	93.38
Transformer	N/A	96.43

the proposed models, showing significant improvements in performance, particularly with the Transformer model.

5 Conclusion

This study developed and evaluated subject-independent and subject-dependent EEG inner speech classification models using LSTM, BLSTM, and Transformer architectures across two datasets. The results demonstrate that subject-dependent models consistently outperform subject-independent ones due to the unique nature of each individual's EEG signals, which makes generalization across subjects challenging.

The Transformer model emerged as the best-performing architecture on both datasets, with preprocessing quality significantly influencing the results. Additionally, we explored ensemble learning techniques, such as voting and bagging, which showed potential in leveraging the strengths of each model. Although ensemble methods were not fully integrated into the system, they demonstrated promise in improving classification performance by balancing contributions from different models.

Subject-independent models, while performing reasonably within the same dataset, faced significant accuracy declines when tested on unseen subjects.The t-SNE visualizations reinforced the need for subject-specific models by highlighting the distinctiveness of EEG patterns across individuals.

In conclusion, subject-dependent models provide superior classification performance for EEG data, and while ensemble learning was not central to our system, it offered a complementary approach to enhance accuracy. Future research could explore hybrid or advanced ensemble strategies.

6 Declarations

6.1 Abbreviations:

CNN: Convolutional Neural Network

DL: Deep Learning

EEG: Electroencephalography

BCI: Brain Computer Interface

BLSTM: Bidirectional Long Short Term Memory

t-SNE: t-distributed Stochastic Neighbor Embedding

LSTM: Long Short-Term Memory

SVM: Support Vector Machine

KNN: K-Nearest Neighbors

CSP: Common Spatial Pattern

6.2 Ethics approval and consent to participate:

- Not applicable

6.3 Funding:

- Not applicable

6.4 Availability of data and materials:

- The dataset analyzed during the current study is publicly available as the Thinking Out Loud and BCI 2a datasets.

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6.6 Competing Interests:

- The authors declare that they have no competing interests.

6.7 Acknowledgements:

- Not applicable

6.8 Authors' contributions

- Dr. A. Shahina : conceptualized the study, designed the methodology, performed data preprocessing and model development, conducted the experiments, and analyzed the results. wrote the original draft of the manuscript.
- Mythreya & Mitual & Likith : Performed data preprocessing and model development,
- Dr. A. Nayeemulla khan: supervised the research, contributed to methodological refinement and neuroscientific interpretation of the results, and provided overall academic oversight. critically revised the manuscript and approved the final version.

  All authors read and approved the final manuscript.

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